

Thomas VAUDESCAL

Quantitative Developer | Quantitative Researcher

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Profile

Quantitative developer and researcher building research-to-production trading systems in **Python, Numba and Rust**: an event-driven options backtester, an exotics pricing engine with Greeks via AAD, and a **live systematic SPX options strategy**. MSc in Financial Engineering, HEC Montréal (GPA **3.88/4.3**, thesis graded **A+**). Prior roles at National Bank Financial Markets (equity derivatives R&D) and Deloitte (model validation).

Education

HEC Montréal | MSc in Financial Engineering | GPA 3.88/4.3

Montréal | August 2022 - December 2024

- **Thesis, A+ with commendation from the examination committee**: Gamma Exposure and Intraday Volatility Dynamics in the SPX Options Market, later applied to a live systematic options strategy.
- **Awards**: four scholarships of excellence (Deloitte M.Sc., Alpha Fixe Capital, Monty I and II), 2nd place at the HEC Montréal Trading Competition, National Bank challenge winning team at ConUHacks VIII (2024), top 10 of 66 teams at the McGill FIAM Portfolio Management Hackathon.
- **Leadership**: Vice President Research at the Trading Club (team of 4). R&D Analyst at the BNI-HEC Investment Fund (\$5M AUM).

HEC Montréal | BBA in Mathematics, Finance & Economics

Montréal | 2017 - 2022

- **GPA**: 3.8/4.3 in specialization courses. Minor in Mathematics at Université de Montréal (probability, statistics, stochastic processes, real analysis, C/C++ programming).

Professional Experience

Synerqo | Founder & Quantitative Researcher

Montréal | May 2025 - Present

- Built an in-house **event-driven options backtesting engine in Python and Numba**, modeling spreads, slippage, commissions, margin and assignment, and validating strategies against QuantConnect before live deployment. Runs a backtest over one month of SPX intraday option chains (3.7M rows) in about 40 ms, 4x to 7x faster than the same code with Numba disabled, with identical results.
- Wrote **from scratch an exotic options and structured products pricing engine** with Monte Carlo and PDE solvers, Greeks via **Adjoint Algorithmic Differentiation (AAD)** on JAX, and Heston, SABR and SVI calibration. Greeks cost about 3x one price (GBM, 500k paths) and run 2.9x faster than bump-and-reprice, within 0.25% of Black-Scholes.
- Developed **Rust execution components** for market-data ingestion, quote updates and order routing, and profiled their latency against the Python research stack.
- Designed, backtested and **deployed a systematic SPX options strategy to live trading**, turning MSc research on dealer gamma exposure into a production signal and execution workflow.
- Prototyped systematic options market making with skew-aware quoting, inventory-based price adjustments and microstructure-based adverse-selection filters.

Deloitte | Quantitative Analyst

Montréal | October 2024 - June 2025

- Developed and validated **credit risk models (PD, LGD, EAD)** for major Canadian financial institutions. Led independent validation through statistical backtesting, sensitivity analysis and stress testing.
- Built **automated monitoring and validation pipelines in Python**, reducing manual review time by **40%**.

National Bank Financial Markets | Quantitative Developer Intern

Montréal | May 2023 - December 2023

- In Global Equity Derivatives R&D, developed and deployed low-volatility and momentum **SmartBeta strategies covering \$800M in AUM** within a \$7B structured products division, with constrained portfolio optimization (cvxpy).
- Built an **end-to-end Python backtesting framework**: signal generation, portfolio construction, transaction costs, risk analytics and performance attribution.
- Delivered **\$1M in annual cost savings** by migrating strategies from an external vendor to internal production systems.

Videns Analytics | Data Analyst

Montréal | April 2020 - June 2021

- Built predictive models and reporting for insurance clients. Designed a **Python and machine learning curriculum** adopted by UQAC.

Projects

Quant Paper Lab | github.com/Thomas4390/quant-paper-lab

Open source | 2026

- Landmark finance papers reproduced on public data with interactive Streamlit figures (Jegadeesh and Titman 1993: published 1.31% monthly momentum return). A test cuts network access to enforce offline computation. Diagnosed a native pyarrow segfault.

FX Strategies | github.com/Thomas4390/fx_strategies

Open source | 2026

- Research framework for intraday and daily FX strategies on VectorBT Pro: parameter grids, purged walk-forward cross-validation and Numba metric kernels. 46 tests, including numerical-equivalence checks (rtol 1e-10) against frozen baselines.

Ganesha | ganesha.synerqo.com

Synerqo | Production

- Production RAG assistant for professional services firms (Python, FastAPI, Qdrant, Next.js): retrieval, reranking, answers linked to their sources, and an LLM evaluation framework with claim-level evidence checks.

Technical Skills & Interests

Programming	Python (NumPy, Numba, pandas, SciPy, statsmodels, scikit-learn, PyTorch, cvxpy, QuantLib), C++ , Rust , SQL, R.
Computer Science	Data structures and algorithms, object-oriented design, event-driven systems, latency profiling, optimization.
Quant Finance	Derivatives pricing, AAD Greeks, Monte Carlo, PDE, Heston/SABR/SVI, backtesting, portfolio optimization.
Tools	Git, Linux, Docker, AWS, Bloomberg Terminal & API, REST and WebSocket APIs.
Languages	French (native), English (professional proficiency).
Interests	Chess (2000 Elo, Chess.com rapid) , marathon running, mountain hiking, indoor climbing, scuba diving.